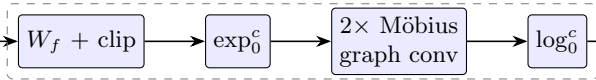
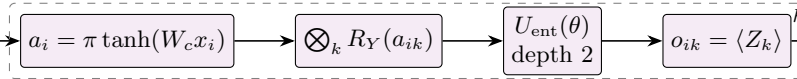


hyperbolic branch (geo), learnable c

geometry $\in \{\text{hyperbolic, euclidean (NEMA-E), off}\}$



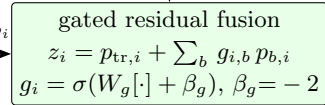
node features $x_i \in \mathbb{R}^F$,
graph $G = (V, E)$



quantum_mode $\in \{\text{pqc, frozen_random } (\theta \text{ fixed: NEMA-R), surrogate (MLP: NEMA-C), off}\}$



classical bypass (trunk)



fusion $\in \{\text{residual, softmax}\}$

head W_h

logits $\hat{y}_i$

per-branch aux heads A_b : deep supervision, weight $w(t) \downarrow 0$

per-branch gradient telemetry + gate $\log g_{i,b}$ (H3, H4)